Deterministic Optimization

Large-Scale Optimization

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Primal-Dual Relationship:
Constraint Generation

Large-Scale Optimization

Learning Objectives for this lesson

- Explore the primal-dual perspective of column generation
- Design algorithm to solve large-scale optimization modeling with many constraints

Dual of Cutting Stock Problem

Item	Patterns												Demand
w_i	1	2	3	4	5	6	7	8	9	10	11	....	N	b_i
25	4	3	2	2	2	1	1	1	1	0	0	....	a_1	150
35	0	0	0	1	0	2	0	1	0	1	0	....	a_2	200
45	0	0	0	0	1	0	1	0	0	0	1	....	a_3	300

$$
\begin{aligned}
\min \quad & \sum_{j=1}^{N} x_j \\
\text{s.t.} \quad & \sum_{j=1}^{N} \boldsymbol{A}_j x_j = \boldsymbol{b} \\
& x_j \geq 0, \quad \forall j = 1, \ldots, N.
\end{aligned}
$$

Dual

$$
\begin{aligned}
\max \quad & 150y_1 + 200y_2 + 300y_3 \\
\text{s.t.} \quad & 4y_1 \leq 1 \\
& 3y_1 \leq 1 \\
& 2y_1 \leq 1 \\
& 2y_1 + y_2 \leq 1 \\
& 2y_1 + y_3 \leq 1 \\
& y_1 + 2y_2 \leq 1 \\
& \ldots
\end{aligned}
$$

Dual Problem has 3 variables but a large number of constraints

Primal-Dual Relationship

$$
\begin{aligned}
(MP) \quad Z = \min \quad & \sum_{j=1}^{N} c_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^{N} a_{ij} x_j = b_i, \quad \forall i = 1, \ldots, m \\
& (\text{Equivalently: } \sum_{j=1}^{N} \boldsymbol{A}_j x_j = \boldsymbol{b}) \\
& x_j \geq 0, \quad \forall i = 1, \ldots, N,
\end{aligned}
$$

$$
\begin{aligned}
(RMP) Z_R = \min \quad & \sum_{j \in I} c_j x_j \\
\text{s.t.} \quad & \sum_{j \in I} a_{ij} x_j = b_i, \quad \forall i = 1, \ldots, m \\
& (\text{Equivalently: } \sum_{j \in I} \boldsymbol{A}_j x_j = \boldsymbol{b}) \\
& x_j \geq 0, \quad \forall j \in I.
\end{aligned}
$$

$\downarrow$ Dual

$$
\begin{aligned}
(DMP) \quad Z_D = \max \quad & \sum_{i=1}^{m} b_i y_i \\
\text{s.t.} \quad & \sum_{i=1}^{m} a_{ij} y_i \leq c_j, \quad \forall j = 1, \ldots, N \\
& (\text{Equivalently: } \boldsymbol{y}^T \boldsymbol{A}_j \leq c_j, \ \forall j = 1, \ldots, N)
\end{aligned}
$$

$\downarrow$ Dual

$$
\begin{aligned}
(DRMP) Z_{DR} = \max \quad & \sum_{i=1}^{m} b_i y_i \\
\text{s.t.} \quad & \sum_{i=1}^{m} a_{ij} y_i \leq c_j, \quad \forall j \in I. \\
& (\text{Equivalently: } \boldsymbol{y}^T \boldsymbol{A}_j \leq c_j, \quad \forall j \in I.)
\end{aligned}
$$

DMP (DRMP) is Dual of MP (RMP)

- There is a complete symmetry:
    - Dual Master Problem (DMP) is the dual of primal P
    - Dual Restricted Master Program (DRMP) is the dual of primal RMP
    - Dual Restricted Master Program (DRMP) is also the restricted version of the Dual Master Problem (DMP)

DMP has Many Constraints

- DMP is given as

$$
\begin{aligned}
(MP) \quad Z = \min \quad & \sum_{j=1}^{N} c_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^{N} a_{ij} x_j = b_i, \quad \forall i = 1, \dots, m \\
& (\text{Equivalently: } \sum_{j=1}^{N} \boldsymbol{A}_j x_j = \boldsymbol{b}) \\
& x_j \geq 0, \quad \forall i = 1, \dots, N, \\
& \qquad \downarrow \text{Dual} \\
(DMP) \quad Z_D = \max \quad & \sum_{i=1}^{m} b_i y_i \\
\text{s.t.} \quad & \boxed{\sum_{i=1}^{m} a_{ij} y_i \leq c_j, \quad \forall j = 1, \dots, N} \\
& (\text{Equivalently: } \boldsymbol{y}^T \boldsymbol{A}_j \leq c_j, \ \forall j = 1, \dots, N)
\end{aligned}
$$

Dual Master Problem has
a small number of variables m
but a large number of constraints N

Constraint Generation

- To solve an optimization problem with many constraints, we can use a similar algorithmic idea as column generation:
    1. Start with a reduced master problem with a subset of constraints
    2. Solve the reduced master problem with optimal solution x^*
    3. Check if all the constraints (especially those not in the reduced master problem) are satisfied at x^*
        a. If yes, then the current solution x^* is optimal for master problem
        b. Otherwise, find a constraint violated by x^*, add it to reduced master problem. Go to Step 2.
- This algorithm is called **Constraint Generation**

Summary

- We learned:
  - The primal-dual relationship that links an optimization problem with many variables (columns) to an optimization problem with many constraints (rows)
  - The constraint generation algorithm